Retrieval-Augmented Generation (RAG)

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July 21, 2026

The problem we're solving

WHY WE NEED RAG

“Does vitamin D supplementation reduce the risk of respiratory infections?”

The answer has to be:

- ✓ Correct
- ✓ Grounded in published evidence
- ✓ Cites its sources

A plain LLM isn't enough

It will happily answer — **sometimes correctly, sometimes with a confidently invented citation.**

Unacceptable wherever being wrong is expensive:

medicine **law** **finance**

RAG changes the question we ask the model

THE CORE IDEA

INSTEAD OF

“What do you know about X?”

tests the model's memory

→

WE ASK

“Here are 5 relevant passages. Using only these, answer X and cite your sources.”

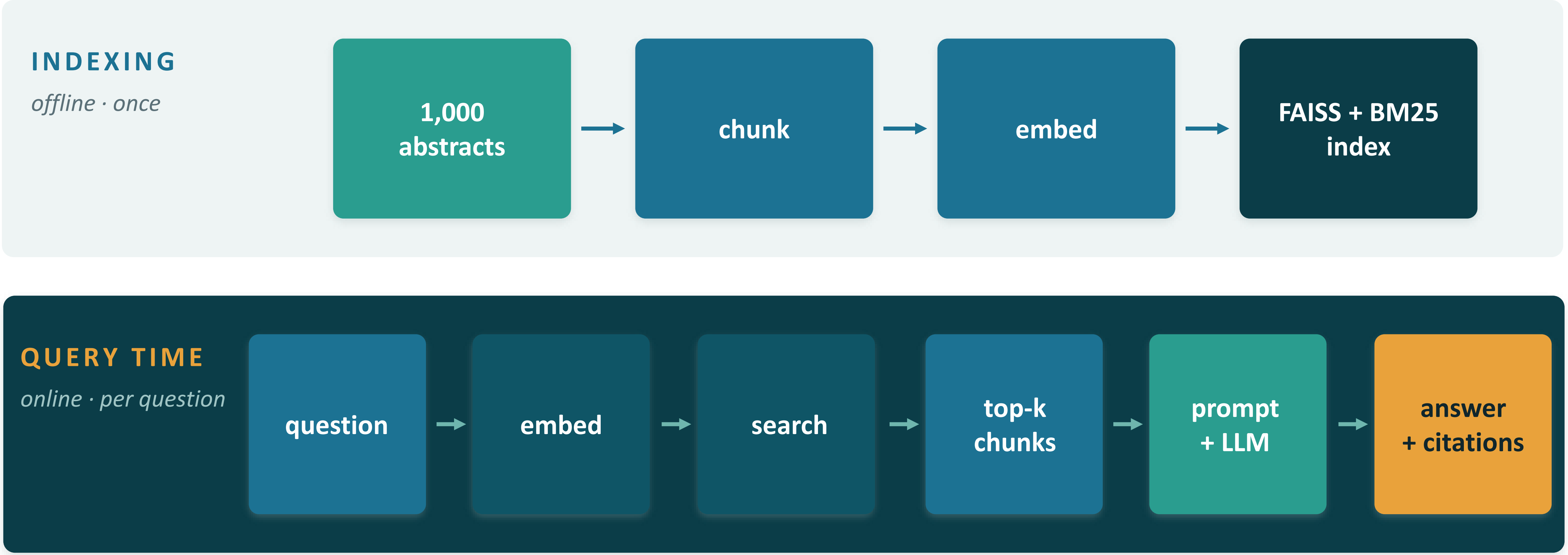
tests the model's reading comprehension

Reading comprehension is far easier than memorisation:

- ✓ Small open models become useful
- ✓ Knowledge updates without retraining
- ✓ Every claim becomes auditable

Index once, then answer many

THE ARCHITECTURE



The dataset: PubMedQA

HANDS-ON

1,000 real biomedical questions

Each row gives us:

- the question
- the source abstract that answers it
- an expert-written long answer
- a yes / no / maybe verdict

Ground truth at retrieval and generation, allowing us to measure both

1 **RETRIEVAL**

We know which document answers each question.

→ measure **Recall@k · MRR@k**

2 **GENERATION**

We know the correct yes / no / maybe verdict.

→ measure **Accuracy · F1 vs. the reference answer**

The next 60+ minutes

ROADMAP

- 0 Setup
- 1 The problem & the corpus
- 2 Chunking (indexing · part 1)
- 3 Embeddings & the vector index (indexing · part 2)
- 4 Retrieval: dense · lexical · hybrid + evaluation
- 5 Prompting & generation
- 6 Evaluating responses
- 7 Failure modes & the production stack